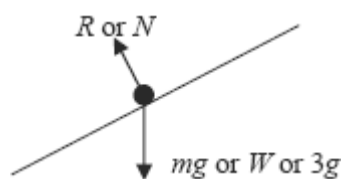


Mark schemes

Q1.

(a) (i)



Correct diagram with arrows and labels

B1

1

(ii) $3a = 3g \sin 30^\circ$

Two term equation of motion

M1

$$a = g \sin 30^\circ = 4.9 \text{ m s}^{-2}$$

AG Correct acceleration from correct working (Allow $a = g \sin 30^\circ$)

A1

2

(b) (i) $5 = \frac{1}{2} a \times 2^2$

Constant acceleration equation with $u = 0$

M1

$$a = 2.5 \text{ m s}^{-2}$$

*AG Correct answer from correct working.
(Use of $v = 5$ must be justified)*

A1

2

(ii) $3 \times 2.5 = 3g \sin 30^\circ - F$

Three term equation of motion

M1

Correct equation

A1

$$F = 3g \sin 30^\circ - 7.5$$

$$= 7.20 \text{ N (to 3 sf)}$$

Correct F

Accept 7.2 N

A1

3

(iii) $R = 3g \cos 30^\circ (= 25.46)$

Resolving perpendicular to the slope to find R

M1

Correct R

A1

$$7.2 = \mu \times 3g \cos 30^\circ$$

Use of $F = \mu R$

M1

Correct expression

A1F

$$\mu = \frac{7.2}{3g \cos 30^\circ} = 0.283$$

Correct μ

Accept 0.282

(Follow through from incorrect F from above, but not an incorrect R)

A1F

5

- (iv) Reduce a , as the air resistance would reduce the magnitude of the resultant force or because the air resistance increases as the velocity increases towards its terminal value

Reduces

B1

Explanation

Second B1 dependent on the first B1 mark

B1

2

[15]

Q2.

(a) $\mathbf{u} = 5\mathbf{i}$ or $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

Correct velocity

B1

1

(b) $\mathbf{v} = 5\mathbf{i} + (-0.2\mathbf{i} + 0.25\mathbf{j})t$

Use of constant acceleration equation,
with $\mathbf{u}$ and $\mathbf{a}$ not zero

M1

Correct velocity
M1A0 for using $5\mathbf{j}$ or just 5

A1

OR

$$\mathbf{v} = \begin{bmatrix} 5 - 0.2t \\ 0.25t \end{bmatrix}$$

2

(c) $5 - 0.2t = 0$

Easterly component zero

M1

Correct equation

A1

$$t = \frac{5}{0.2} = 25 \text{ seconds}$$

Correct t

A1

3

(d) $\mathbf{r} = 5\mathbf{i} \times 25 + \frac{1}{2} (-0.2\mathbf{i} + 0.25\mathbf{j}) \times 25^2$

Use of constant acceleration equation with
 t from part (c)

M1

Correct expression based on t from part (c)

A1F

$$= 62.5\mathbf{i} + 78.125\mathbf{j}$$

Correct simplification CAO

A1

$$\theta = \tan^{-1} \left(\frac{62.5}{78.125} \right)$$

Using *tan* to find the angle

dM1

Correct expression based on *t* from part (c),
with correct two values (either way)

A1F

$$= 038.7^\circ$$

Correct angle
Accept 38.6° or 039°

A1

OR

$$\mathbf{r} = \frac{1}{2} (5\mathbf{i} + 6.25\mathbf{j}) \times 25$$

(M1)(A1F)(A1)

$$\theta = \tan^{-1} \left(\frac{5}{6.25} \right) = 038.7^\circ$$

(dM1)(A1F)(A1)

6

[12]

Q3.

(a) $v^2 = u^2 + 2as$

$$0 = 10.5^2 - 2 \times 9.8 \times s$$

full method

M1

$$s = 5.625 \text{ metres} \quad (5.63) \quad (5.62)$$

A1

2

(b) (i) $0 = 10.5 - 9.8t$

M1

$$t = 1.07 \text{ sec} \quad (1.0714...)$$

A1

2

(ii) $T = 2.14 \text{ sec}$

B1ft

1

[5]

Q4.

(a) $s = ut + \frac{1}{2} at^2$

$$0 = 2\frac{1}{2}ut - \frac{1}{2}gt^2$$

M1

*Full method required for time
(equations of motion, or standard result)*

A1

$$0 = t \left(2\frac{1}{2}u - \frac{1}{2}gt \right)$$

m1

$$t = \frac{5u}{g}$$

(if $g = 9.8$ used, lose last A1)

A1

4

(b) $0A = 6u \times \frac{5u}{g}$

M1

$$= \frac{30u^2}{g}$$

CAO

A1

2

(c) $\text{speed}^2 = (6u)^2 + \left(2\frac{1}{2}u \right)^2$

M1

$$\text{speed} = 6\frac{1}{2}u$$

CAO

A1
2

(d) Least speed, at top, = $6u$

B1
1

[9]

Q5.

(a) (i) $a = \frac{18 - 12}{10}$

$\pm$

M1

$$a = 0.6 \text{ ms}^{-2}$$

A1
2

(ii) $s_1 = \frac{1}{2} (12 + 18) \times 10$

M1

$$s_1 = 150 \text{ metres}$$

EXAM PAPERS PRACTICE
 ©2025 Exam Papers Practice. All rights reserved.

A1
2

(b) $s_2 = 20 \times 12 = 240 \text{ metres}$

B1
1

(c) $s_3 = \frac{1}{2} (12 + 18) \times 20$

M1

$$= 300$$

PI

A1

$$\text{average speed} = \frac{150 + 240 + 300}{50}$$

M1

$$= 13.8 \text{ms}^{-1}$$

ft one slip

A1ft

4

[9]

Q6.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

A1

2

(b) $t = 0.5, \mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

$$\text{Speed} = \sqrt{(2.5^2 + 6^2)}$$

2 terms

M1

$$\text{Speed} = 6.5 \text{ms}^{-1}$$

ft 2 terms

A1ft

3

[5]

Q7.

(a) (i) $s = ut + \frac{1}{2}at^2$

$$25 = 0 + 4.9t^2$$

full method

M1

$$t = 2.26 \text{ sec} \quad (2.236)(\text{if } g = 10)$$

(2.259)

A1
2

(ii) $v^2 = u^2 + 2as$

$$v^2 = 0 + 2 \times 9.8 \times 25$$

M1

$$v = 22.1 \text{ ms}^{-1} \quad (21.913)$$

A1
2

$$(22.14)$$

- (b) (Time longer) air resistance slows down motion,
links with motion, no contradictions

M1

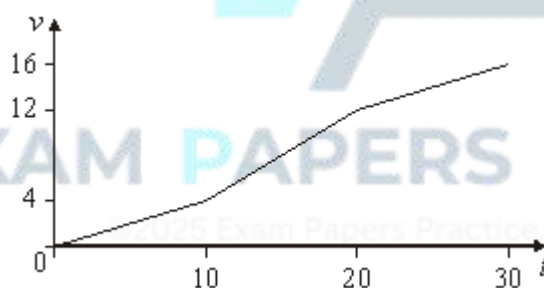
*(or Time less) package large so less
distance to travel*

A1
2

[6]

Q8.

(a) (i)



3 straight lines

B1

correct end points

B1

sensible scales ÷ labelled v/t

B1
3

(ii) $s = \frac{1}{2} \times 10 \times 4 + \frac{1}{2} \times (4 + 12) \times 10 + \frac{1}{2} (12 + 16) \times 10$

M1

Or
 area attempt (M1)
 full method (m1)
 equation correct (A1)
 ft one slip

m1

Or
 equation attempted (M1)
 full method (m1)
 all correct (A1)
 ft one slip

A1

$s = 240$ metres

A1ft

4

(iii) Average speed = $\frac{240}{30}$

M1

$= 0.8 \text{ ms}^{-1}$
 cao

A1

2

(iv) Greatest acceleration = 2nd stage

$= \frac{12 - 4}{10}$

M1

$= 0.8 \text{ ms}^{-2}$
 CAO

A1

2

(b) (i) Less

B1

area below curve < area below line/velocity lower
 no additional incorrect statements

B1

2

(ii) Change in velocity more gradual oe

B1

1

[14]

Q9.

(a) $s = 0 + \frac{1}{2} \times 9.8 \times 4^2$

full method

M1

correct subs, accept ± 9.8

A1

$s = 78.4 \text{ metres}$

CAO (need positive)

A1

3

(b) $v = 0 + 9.8 \times 4$

full method

M1

$v = 39.2 \text{ ms}^{-1}$

CAO (need positive)

A1

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[5]

Q10.

(a) $v = u + at$

$0 = 10 + (-0.8) \times t$

Full method with u, v used correctly, Accept

M1

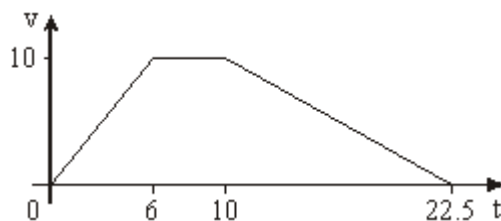
$t = 12.5 \text{ sec}$

CAO, (correct substitution and answer)

A1

2

(b)



each line, straight and correct end points

B1 B1 B1

SC first error in times lose B1, subsequent repeated errors, no further penalty

SC B1 for 3 lines giving correct shape but no values shown axes labelled, v, t

B1

4

(c)
$$\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$$

full correct method (including addition of distances)

M1

correct subs, ft graph if final t = 12.5

A1

$$= 132.5 \text{ metres}$$

ft one slip AWRT 133

A1F

3

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[9]

Q11.

(a) $\mathbf{d} = 3\mathbf{i} - 6\mathbf{j}$

Accept $\pm \mathbf{d}$ or displacement of 3, 6 shown on a diagram

B1

$$3\mathbf{i} - 6\mathbf{j} = (\mathbf{i} - 2\mathbf{j})t$$

Or equivalent method for t

Accept ratio of vectors leading directly to ± 3

M1

$$t = 3$$

CAO

A1

3

(b) (i) $\mathbf{r} = (\mathbf{i} - 2\mathbf{j}) \times 4 + \frac{1}{2} \times 2\mathbf{j} \times 16$

M1

*full method for vector expression giving change in position
for correct subs (give $4\mathbf{i} + 8\mathbf{j}$)*

A1

$+6\mathbf{i} - 4\mathbf{j}$

M1

$= 10\mathbf{i} + 4\mathbf{j}$

*ft slip provided obtain vector expression
($\mathbf{u} = 0$ gives $6\mathbf{i} + 12\mathbf{j}$)*

A1F

4

(ii) $A(3,2) \ C(10,4)$

$\mathbf{d} = 7\mathbf{i} + 2\mathbf{j}$

*Attempt to find vector **AC** or **CA**
(using candidate's C)*

M1

$|\mathbf{d}| = \sqrt{7^2 + 2^2}$

$AC = \sqrt{53} = 7.28$

ft $\mathbf{d}$ provided two non-zero components

Accept $\sqrt{53}$

A1F

2

[9]

Q12.

(a) $57 = 24 \cos 40^\circ \times t$

component attempted and acceleration = 0

		M1	
	<i>all correct</i>		
		A1	
	$t = 3.10 \text{ sec}$		
	CAO		
		A1	
			3
(b)	$h = 24 \sin 40^\circ \times 3.1 - \frac{1}{2} \times 9.8 \times 3.1^2$		
	<i>component attempted & acceleration = 9.8</i>		
		M1	
	<i>all correct</i>		
		A1	
	$h = 0.734 \text{ m}$		
	<i>ft one slip e.g. + 9.8 used</i>		
	<i>Accept 2 SF answer, AWR 0.71-0.74</i>		
		A1F	
			3
(c)	(i) horizontal, $u = 24 \cos 40^\circ = 18.39 \text{ ms}^{-1}$		
	<i>Seen anywhere in (c), accept 18.4</i>		
		B1	
	vertical, $v = 24 \sin 40^\circ - 9.8 \times 3.1$		
	<i>component attempted & acceleration = 9.8</i>		
		M1	
	$v = -14.95 \text{ ms}^{-1}$		
	<i>(accept -15.0)</i>		
		A1	
	$V = \sqrt{(18.39^2 + (-14.95)^2)}$		
	<i>Use of candidate's u and 'new' v</i>		
		M1	
	$V = 23.7 \text{ ms}^{-1}$		
	<i>ft use of candidate's u and v</i>		

A1F

5

(ii) $\tan \theta = \frac{14.95}{18.39}$

Use of candidate's u and v (new)

Accept inverted ratio

M1

$\theta = 39.1^\circ \text{ or } 39.2^\circ, 140.8^\circ, 140.9^\circ \text{ accept } \pm$

ft use of candidate's u and v and V

A1

2

[13]

Q13.

(a) $s = 0 + \frac{1}{2} \times 9.8 \times 4^2$

Full method

M1

Correct subs, accept ± 9.8

A1

$s = 78.4 \text{ metres}$

CAO (need positive)

A1

3

(b) Average speed $= \frac{78.4}{4}$

Also accept full method with use of velocities at

$t = 0$ and 4, or at $t = 2$

M1

$= 19.6 \text{ ms}^{-1}$

FT distance

A1F

2

(c) Only force acting is weight

*Acc resistance forces negligible or ignored,
(not friction, or air friction)*

B1

1

[6]

Q14.

(a) $v = u + at$

$$0 = 10 + (-0.8) \times t$$

Full method with u, v used correctly

Accept ± 0.8

M1

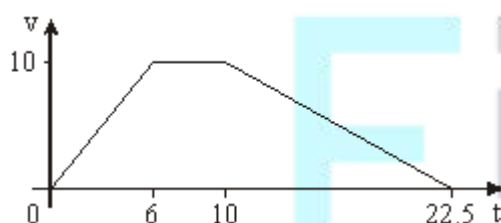
$$t = 12.5 \text{ sec}$$

CAO (correct subs and answer)

A1

2

(b)



each line, straight and correct end points

B1 B1 B1

SC: B1 for 3 lines giving correct shape but no values shown

SC: first error in labelling times loses B1, repeated errors no further penalty

axes labelled v, t

B1

4

(c) $\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$

Full correct method

M1

Correct subs, FT graph if final t = 12.5

A1F

$$= 132.5 \text{ metres}$$

FT one slip, AWRT 133

A1F

3

- (d) Acceleration unlikely to:
change so abruptly or be constant
or velocity unlikely to be constant

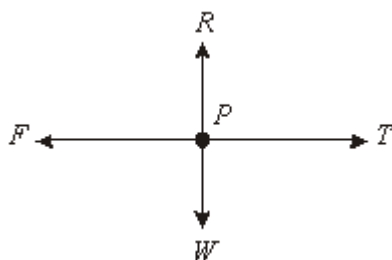
B1

1

[10]

Q15.

- (a) (i)



Accept mg , $0.4g$ or 3.92 for weight
Arrows and labels needed

B1

1

(ii) $F = 0.5 \times (0.4 \times 9.8)$

Need to see 0.4×9.8 or 3.92 used

M1

$F = 1.96\text{N}$

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

A1

2

(b) $T - 1.96 = 1.4a$
 $0.3g - T = 0.3a$

M1A1

Consistent reversal of signs in both equations
4 marks; reversal of signs in one equation,
M1 A1 M1 A0

M1A1

$a = 1.4\text{ms}^{-2}$

Sign change needs justification (whole string:
equation, $0.3g - 1.96 = 0.7a$ M1A1 $a = 1.4$ A1)
max 3/5

A1

5

(c) $v = 14 \times 3$
 $v = 4.2 \text{ms}^{-2}$

Full method

M1

CAO

A1

2

(d) P : Friction will cause
 speed to decrease

M1

Accept decelerate or comes to rest

A1

Q : Gravity will cause
 speed to increase

M1

Accept accelerate

A1

4

[14]

Q16.

(a) $y = -\frac{1}{2}gt^2 + 20\sin 30t$

M1A1

$$x = 20 \cos 30t$$

M1

$$t = \frac{x}{20 \cos 30}$$

A1

$$y = -\frac{1}{2}g \frac{x^2}{400 \cos^2 30} + 20 \sin 30 \frac{x}{20 \cos 30}$$

M1

$$y = x \tan 30 - \frac{gx^2}{800 \cos^2 30^\circ}$$

AG

A1

6

(b) $2.5 = x \tan 30 - \frac{9.8x^2}{800 \cos^2 30}$

$$9.8x^2 - 346x + 1500 = 0$$

substituting and tidying up

M1A1

$$x = \frac{346 \pm \sqrt{119716 - 58800}}{19.6}$$

M1

=30.3 (or 30.2) & 5.06 (or 5.05)
 answer: 30.3m (or 30.2m)
 at least 3 s.f. required

A1F

4

(c) no air resistance,

B1

the ball is a particle

B1

2

etc.

[12]

Q17.

(a) (i) $40 = 12 + 100a$

*Use of a constant acceleration equation to
 form equation for a*

M1

$$a = \frac{40 - 12}{100} = 0.28 \text{ ms}^{-2} \quad \text{AG}$$

AG; correct answer from correct working

A1

2

(ii) $s = \frac{1}{2}(12 + 40) \times 100$

Expression for distance, using $t = 100$

M1

$= 2600\text{m}$

Correct final distance

A1

2

(c) $F - 40000 = 200 \times 1000 \times 0.28$

Three term equation of motion

M1

$F = 40000 + 56000 = 96000 \text{ N}$

Correct equation

A1

Correct force

A1

3

[7]

Q18.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

Correct $\mathbf{a}$

A1

3

(b) $\mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$

Use of constant acceleration equation for position

M1

$$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$$

Correct i component

A1

Correct j component

A1ft

3

ft incorrect acceleration

(c) (i) $\mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j}$
 $= 0\mathbf{i} + 20\mathbf{j}$

Substituting t = 20 into their expression for r

M1

so due north of origin

Correct conclusion from correct working

A1

2

(c) (ii) $\mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$

$$|\mathbf{v}(20)| = \sqrt{4^2 + 3^2} = 5 \text{ ms}^{-1}$$

Finding velocity at t = 20

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft

4

ft incorrect acceleration

[12]

Q19.

(a) $s = \frac{1}{2} \times 15 \times 20 = 150$

M1: Finding length of first stage

M1: Finding length of second stage

A1: Both distances correct

M1

$$s = \frac{1}{2} \times 15 \times 80 = 600$$

M1A1

$$s = 600 + 150 = 750 \text{ m}$$

A1: Correct total distance

A1

4

(b) (i) $t = \frac{750}{15} = 50 \text{ s}$

B1: Correct time or their distance
correctly divided by 15

B1ft

1

(ii) Delay = $120 - 50 = 70 \text{ s}$

B1: Correct time of their previous time
correctly subtracted from 120 to give a
positive answer

B1ft

1

(c) $a = \frac{15}{80} = \frac{3}{16} = 0.1875 \text{ ms}^{-2}$

M1: Finding acceleration

A1: Correct acceleration

M1A1

$$F = 500000 \times 0.1875 = 93800 \text{ N (to 3sf)}$$

M1: Use of $F = ma$

A1: Correct force

M1A1

4

[10]

Q20.

(a) $9g - T = 9a$

M1: Equation for one particle

A1: Correct equation

M1A1

$$T - 5g = 5a$$

M1: Equation for other particle
A1: Correct equation

M1A1

$$4g = 14a$$

$$a = \frac{4g}{14} = 2.8 \text{ ms}^{-2}$$

A1: Correct a from correct working
AG

A1

5

(b) $T - 5g = 5 \times 2.8$

M1: Substituting acceleration to find T

M1

$$T = 63 \text{ N}$$

A1: Correct tension

A1

2

(c) $s = \left(\frac{1}{2} \times 2.8 \times 0.5^2 \right)$

M1: Constant acceleration equation with
 $u = 0$ and $a \neq g$ to find s . Allow $\pm$ answers.
A1: Correct equation

M1A1

$$= 0.35 \text{ m}$$

A1: Correct distance

A1

$$\text{Total} = 2 \times 0.35 = 0.7 \text{ m}$$

A1: Doubling their distance to get total
distance apart

A1ft

4

[11]

Q21.

(a) $2 = \frac{1}{2} \times 9.8t^2$

M1: Vertical equation with two terms including a 2
A1: Correct equation

M1A1

AG

$$t = \sqrt{\frac{4}{9.8}} = 0.639 \text{ s}$$

dM1: Solving for t
A1: Correct t from correct working

dM1A1

4

(b) $s = 30 \sqrt{\frac{4}{9.8}} = 19.2 \text{ m}$

M1: Use of $30 \times \text{time}$
A1: Correct distance

M1A1

2

- (c) (i) Horizontal remains constant
B1: Correct description

B1

- (ii) Vertical increases in magnitude
B1: Correct description

B1

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[8]

Q22.

(a) $\mathbf{v} = 4\mathbf{j} + (3\mathbf{i} - 5\mathbf{j})t$

M1: Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ and $\mathbf{u} \neq 0$ or Integration
A1: Correct expression

M1A1

2

(b) $\mathbf{v} = 3t\mathbf{i} + (4 - 5t)\mathbf{j}$

M1: j component of velocity equal to zero

M1

$$4 - 5t = 0$$

$$t = 0.8 \text{ s}$$

A1: Correct t

A1

2

(c) $\mathbf{v} = 12\mathbf{i} - 16\mathbf{j}$

M1: Finding velocity when $t = 4$

M1

$$v = \sqrt{12^2 + 16^2} = 20 \quad \mathbf{AG}$$

A1: Correct velocity

A1

dM1: Finding the magnitude

dM1

A1: Correct speed from correct working

A1

4

[8]

Q23.

(a) Ball is a particle/no spin

B1: One assumption

B1

No air resistance/Only gravity or weight

B1: Second assumption

B1

2

(b) (i) $24.5t - 4.9t^2 = 0$

M1: Equation for vertical motion with height zero

M1

$$\left(t = 0 \text{ or } t = \frac{24.5}{4.9} = 5\text{s} \right) \quad \mathbf{AG}$$

A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

(b) (ii) $R = 10 \times 5 = 50 \text{ m}$

M1: Use of horizontal component of velocity to find the range

M1

A1: Correct range

A1

2

(c) $20 = 10 t$

M1: Horizontal equation

M1

$t = 2$

A1: Time to reach wall

A1

$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4 \text{ m}$

dM1: Vertical equation for height with $u = 24.5$ and a negative acceleration

dM1

A1: Correct height

A1

4

- (d) No change as acceleration and initial velocity do not change with the mass

B1: No change

B1

B1: Explanation

B1

2

[14]

Q24.

(a) $20 = \frac{1}{2} (3 + V)4$

Full method, ($a = 1$)

M1

$$V = 7$$

A1

2

(b) (i) $S = 2V (=14)$

Alternative: Time = $T - 4$

M1

$$35 - S = \frac{1}{2} \times t \times V$$

Full method $35 = \frac{1}{2}(T - 4 + 2) \times V$

M1

$$21 = \frac{7}{2}t_2$$

Correct subs $35 = \frac{1}{2}(T - 2) 7$

A1

$$t = 6 \quad t = 12$$

ft one slip $T = 12$

A1F

4

EXAM PAPERS PRACTICE

(ii) $f = \pm \frac{7}{6} = 1.17 \quad (1.1666)$

Accept $\pm$ ft time

M1A1F

2

(c) Average speed = $\frac{\text{Total distance}}{\text{time}} = \frac{55}{12}$

Attempt at full area for distance

M1

$$= 4 \frac{7}{12} \quad (4.5833)$$

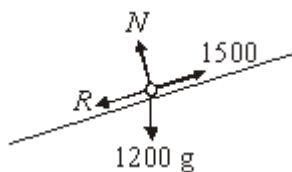
ft time

A1F

2

Q25.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$
M1: 3 terms with component attempted
($\alpha = 4.096$) – 1 each error

M1A2

$R = 660$
AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $\frac{1500}{1200}$
Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1

2

(ii) $0 = 15 + (-1.25) t$
velocities correct, but accept $\pm$ acceleration

M1

$$t = 27.3 \text{ sec} \quad (27.27)$$

ft one slip

A1F

2

[11]

Q26.

(a) $5mg - T = 5m \frac{g}{4}$

3 terms, recognisable, accept m missing or g missing once

M1

all correct & algebraic

A1

$$T = 15m \frac{g}{4} = 3.75 mg$$

ft 9.8 used, one g missing, m missing, sign error provided T positive

A1F

3

(b) $T - kmg = km \frac{g}{4}$

as in (a)

M1A1

$$k = 3$$

CAO
following fully correct work in (a) & (b)

A1

3

(c) $\text{Force} = 2T = \frac{15mg}{2} = 7.5 mg$

B1F

1

(d) $t = \frac{2}{3}$

$$x = 0 + \frac{1}{2} \times \frac{g}{4} \times \frac{4}{9} \text{ accept } \left(\frac{2}{3}\right)^2$$

method for x , and $\frac{g}{4}$ used for acceleration

M1

accept 9.8 substituted

A1

$$d = 2x = \frac{g}{9}$$

Fully correct, in terms of g

A1

3

[10]

Q27.

(a) (i) $V = 0 + 3 \times 0.8$

M1

$V = 2.4$
AG

A1

2

(ii) $S = \frac{1}{2} (5 + 8) \times 2.4$

Method for total area

M1

A1

$S = 15.6\text{m}$
ft slip

A1F

3

(b) $\frac{1}{2} \times T \times 2.4 = 4.8$

Full method for time

M1

$$T = 4$$

A1

$$\text{Average speed} = \frac{15.6 + 4.8}{8 + 4}$$

Total distance / total time

M1

$$= 1.7\text{ms}^{-1}$$

ft one error in S or in T

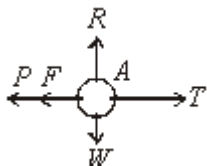
A1F

4

[9]

Q28.

(a) (i)



For any 3 forces correct

B1

*For the rest correct, and no extras
(accept F in either direction, but not μR)*

B1

2

(ii) $R = 0.2 \times g$

$$F = 0.5 \times R$$

For equations for both R and F

M1

$$F = 0.98\text{N}$$

AG

A1

2

(iii) $T = 0.3 \times 9.8 = 2.94\text{N}$

M1A1

2

(iv) $P + F = T$

M1

least $P = 1.96\text{N}$

ft T provided $P > 0$

A1F

2

(b) (i) $2.94 - T = 0.3a$

M1A1

$T - 0.98 = 0.2a$

M1 either equation

(whole string method: 'correct' equation, M1 A1, answer A1; max 3 / 5)

A1

$1.96 = 0.5a$

m1

$a = 3.92\text{ms}^{-2}$

ft one error

A1F

5

(ii) $0.1 = \frac{1}{2} \times 3.92 \times t^2$

Use of candidate's acceleration, if 9.8

M1

$t = 0.226$ (0.22588)

ft candidate's acceleration (not 9.8) if

$a = 1.96$ used, $t = 0.319$

A1F

2

[15]

Q29.

(a) (i) $x = 14t$

Accept as components of position vector

Accept $x = 15.7 \cos 26.6$

B1

$y = 7t - 4.9t^2$
Use of equation

M1

3

Substitution

A1

Accept g; accept $y = 15.7 \sin 26.6t - \frac{1}{2}gt^2$

(ii) $t = \frac{x}{14} \quad y = 7 \times \frac{x}{14} - 4.9 \times \left(\frac{x}{14}\right)^2$

Use of candidate's coordinates

M1

Substituted; accept unsimplified

A1

$= \frac{x}{2} - 4.9 \frac{x^2}{196}$

$y = \frac{20x - x^2}{40}$

AG; convincingly found

A1

3

(iii) $y = 0 \quad 20x - x^2 = 0$

*Full method for range ($t = 1.4286$ for range,
or $t = 0.7143$ for greatest height and horizontal*

distance then doubled, or $R = \frac{V^2 \sin 2\alpha}{g}$

M1

$x = 20$

AWRT

A1

2

(b) $x = 12, y = 2.4$

Substitution into appropriate equation(s), ($t = 0.857$)

M1

For $y = 2.4$

A1

$\therefore$ under bar

ft provided $y > 0$

Alt: M1 for $x = 10$ subs and followed by comparison,

$H = 2.5$ A1, conclusion A1

A1F

3

(c) Air resistance ignored, ball treated as particle...

B1

1

[12]

Q30.

(a) $v^2 = 5^2 + 2 \times (-1.8) \times 2.5$

3 terms for M1, accept 1.8; A1 all correct

M1A1

$v = \pm 4 \text{ ms}^{-1}$

Both required

A1

3

(b) $v = u + at, \quad -4 = 4 - 1.8t$

M1 for full method for finding the two times at B or their difference

A1F if one positive & one negative time

M1A1F

$t = 40/9$ or $t = 4 \frac{4}{9}$ or 4.44 sec

A1F for completion, including difference of times

A1F

3

Alternatives:

$$s = ut + \frac{1}{2} at^2, \quad 2.5 = 5t - \frac{1}{2} \times 1.8t^2$$

(M1)

$(9t - 5)(t - 5)$

(A1)

$$\text{time difference} = 4 \frac{4}{9}$$

(A1F)

*If time from B to stopping point found,
20/9, M1A1F*

time $\times$ 2, 40/9, A1F

[6]

Q31.

(a) $x = 7t$

B1

$$y = 0 + \frac{1}{2}gt^2$$

Accept $\pm$

M1

$$y = 4.9t^2$$

Accept $\pm$

A1

3

(b) $t = \frac{x}{7}$

*Attempt at substitution, or use of equation of
trajectory with $V = 7$ & $\alpha = 0$*

M1

$$y = 4.9 \left(\frac{x}{7} \right)^2$$

$$y = \frac{x^2}{10}$$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x , accept $\pm$

M1

$$x = 9\text{m}$$

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

*For M1 Accept $\pm$,
for A1 consistency of signs needed*

M1A1

$$u = 7$$

B1

$$\text{speed}^2 = (7^2 + 12.6^2)$$

M1

$$\text{speed} = 14.4 \text{ ms}^{-1}$$

(14.414)

A1F

5

[12]

Q32.

(a) $v^2 = u^2 + 2as$

Use of full method

M1

$$0^2 = 3.5^2 + 2 \times a \times 2.5$$

Correct subs

A1

$$a = -2.45 \text{ mag } 2.45$$

Magnitude required

A1

3

(b) (i) Friction : $0.2 \times 2.45 = 0.49\text{N}$

M1: use of $F = ma$ with ± 2.45

A1: Magnitude required

M1A1

2

(ii) $R = 0.2 \times g$
 $F = \mu R \quad 0.49 = \mu \times 0.2g$
Use of $F = \mu R$ with $R = mg$ substituted

M1

$\mu = 0.25$
f.t. (i) provided μ positive
Use of $F < \mu R$, M1 A0

A1F

2

[7]

Q33.

- (a) $a = \text{change in velocity} / \text{time or}$
used

M1

$v = u + at$
 $a = 2 \text{ m s}^{-2}$

A1

2

- (b) distance = $\frac{1}{2} \times 5 \times 6$
Full method

M1A1F

15 metres

A1F

3

- (c) acceleration not likely to be constant initial acceleration
 unrealistic change in velocity at $t = 3$ will be smoother

B1

1

[6]

Q34.

- (a) (i) $B: T = mg = 0.1 \times 9.8 = 0.98\text{N}$
cao

B1

1

(ii) $A: F = T (= 0.98) \quad R = 0.5 \times 9.8$

B1

$$\mu \times 0.5 \times 9.8 = 0.98$$

Condone inequality

M1

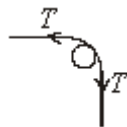
$$\mu = 0.2$$

cao

A1

3

(iii)



Must label T or 0.98 N

B1

1

(iv)



$$R = \sqrt{T^2 + T^2} \quad \text{or } 2T \cos 45^\circ$$

©2025 Exam Papers Practice. All rights reserved.

M1

$$R = 1.39\text{N}$$

A1

2

(b) (i) $A: T - 0.98 = 0.5a$

Either equation for M1

M1A1

$$B: 2 \times 0.1 \times 9.8 - T = 0.2a$$

A1

$$0.98 = 0.7a$$

m1

$$a = 1.4$$

cao

A1

5

(ii) $T = 0.98 + 0.5 \times 1.4, \quad T = 1.68$
ft accuracy error in equation

A1F

1

(iii) $s = ut + \frac{1}{2} at^2$

$$0.7 = 0 + \frac{1}{2} \times 1.4t^2$$

M1

$$t = 1$$

A1

2

[15]

Q35.

(a) (i) vertical: $s = ut + \frac{1}{2} at^2$
 $0 = 7t - 4.9t^2$
Full method

M1

$$0 = t(7 - 4.9t)$$

$$0 = 7 - 4.9t$$

m1

$$t = \frac{10}{7} \text{ sec (1.43)}$$

A1

3

(ii) $OF = 21 \times \frac{10}{7}$

M1

$$= 30 \text{ m}$$

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$$t = \frac{3.5}{4.9}$$

m1

$$FG = 21 \times \frac{3.5}{4.9}$$

M1

$$= 15 \text{ m}$$

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$$OH = 30 + 15 + 7.5$$

M1

$$= 52.5 \text{ m}$$

A1F

3

[12]